MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, POWER UNIT

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-60898 comprise the complete requirements for the Power Unit Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-60898	Circuit Card Assembly, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

FTO554186	Electronic Component Assembly, Power Unit

PACKAGING DATA SHEET

10554186	Packaging of Circuit Card Assembly, Power Unit

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4931

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3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F10554186 and drawings pertaining thereto.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-60898.

3,3 Environmental service conditions. See MIL-C-60898.

3.3.1. Performance at temperature extremes, For the conditions of 3.3.2 of MIL-C-60898, the requirements of 3,4 shall be met with exceptions as specified in the following subparagraphs,

3.3.1.1 Diodes - reverse current. The reverse current shall be as specified in 3.4. 1 except the current between the specified pins shall be 0.1 milliampere or less.

3.3.1.2 Diodes - forward drop, The forward drop voltage shall be as specified in 3.4.2 except the voltage between the specified pins shall be 1.0 ± 0.5 vdc.

3.3.1.3 Resistance. The resistance shall be as specified in 3.4.3 except the resistance tolerance shall be ± 10 percent.

3.3.1.4 Pulse generator high frequency. The high frequency pulse output shall be as specified in 3,4.4.1 except the duty cycle shall not exceed 2.5 milliseconds and the pulse height shall be 10 ± 2 vdc.

3.3.1.5 Pulse generator low frequency. The low frequency pulse output shall be as specified in 3.4.4,2 except the duty cycle shall be not less than 10 milliseconds and the pulse height shall be +14 ± 3 vdc.

3.4 Electrical performance

3.4.1 Diodes, reverse current, With +28.0 ± 0.1 vdc applied between each of the two pins specified in Table I there shall be 0.02 milliampere or less current between those two pins.

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TABLE I

+28.0 ± 0.1 vdc - Polarity	
Pin (+)	Pin (-)
24	26
11	28
3	28
5	28
10	28
7	28
12	28
17	28
18	28
9	28
19	28
5	10
5	2
5	3

3.4.2 Diodes, forward drop. With a current of 1.0 ± 0.1 ampere applies between each of the two pins specified in Table II there shall be 0.9 ± 0.3 vdc between those two pins,

TABLE II

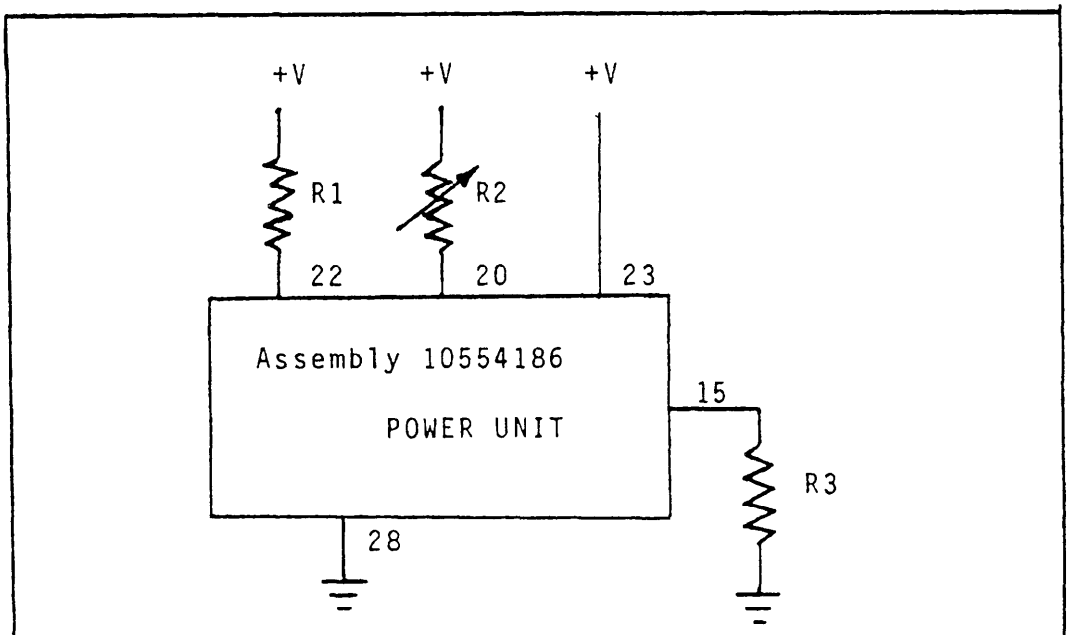
1 Ampere - Polarity	
Pin (+)	Pin (-)
26	24
28	11
28	3
28	5
28	10
28	7
28	12
28	17
28	18
28	9
28	19
10	5
2	5
3	5

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3,4,3 Resistance. The resistance between pin 21 and pin 27 shall be 330 ohms ± 5 percent.

3,4,4 Pulse generator

3,4.4.1 High frequency. With the terminations to the circuit board as shown in Figure 1 (+V = +22.0 ±0.1 vdc and R1 =0.0 ohms) the output shown in Figure 2 shall be present between pin 15 and pin 28. The pulse width shall be obtained by adjusting R2 (reference Figure 1.)



R1 - 0.0 ohms or 2.5 megohms, ± 10 percent, 2 watt.

R3 - 10 kilohms, ± 5 percent, 1/2 watt.

R2 - 1 megohm potentiometer, ± 10 percent, 2 watt.

+V - +22.0 ±0.1 VDC or +28.0 ± 0.1 VDC.

FIGURE I

3.4.4.2 Low frequency. With the terminations to the circuit board as shown in Figure 1 (+V = +28.0 ±0.1 vdc and R1 = 2.5 megohms) the output shown in Figure 3 shall be present between pin 15 and pin 28. The pulse width shall be obtained by adjusting R2 (reference Figure I)

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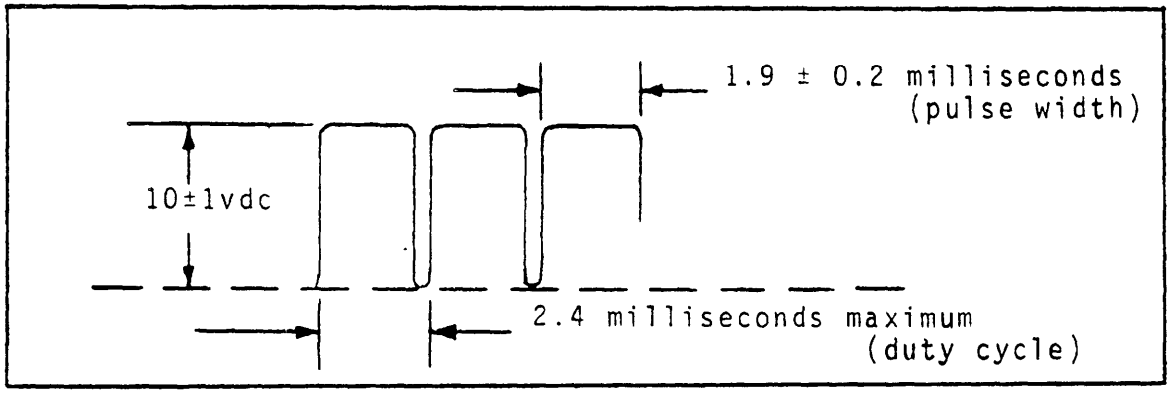


Figure 2

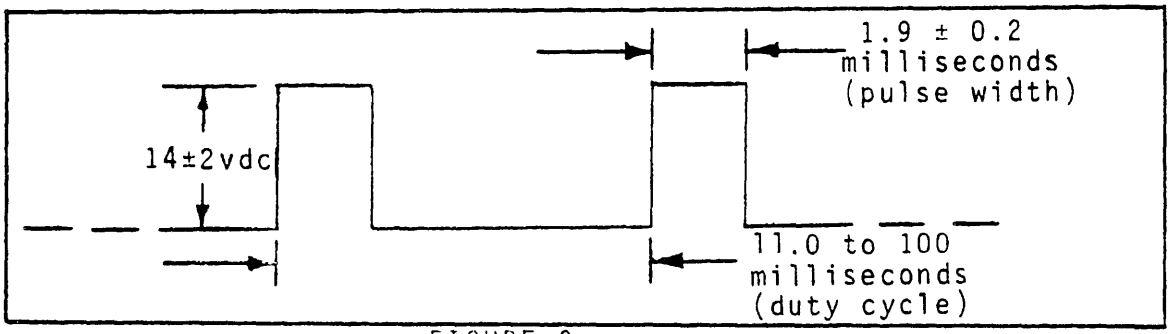


FIGURE 3

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4 QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-60898 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-60898.

4.3 Control test sample. As specified in MIL-C-60898,

4.4 Functional tests. The following characteristics shall be tested 100 percent in accordance with the applicable test method.

Characteristic	Requirement	Procedure
100 Diodes, reverse current	3.4.1	4.5.2.1
101 Diodes, forward drop	3.4.2	4.5.2.2
102 Resistance	3.4.3	4.5.2.3
103 High frequency pulse	3.4.4.1	4.5.2.4.1
104 Low frequency pulse	3.4.4.2	4.5.2.4.2

4.4.1 Inspection equipment. As specified in MIL-C-60898 and 4.4.1.1.

4.4.1.1 Test equipment. In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable test as specified in 4.5:

Item		Description
1.	POWER SUPPLY	0-30 vdc, 0-1.5 ampere, current and voltage adjustable. Ripple less than 100 mv rms.
2.	AMMETER	Accuracy and resolution at .02 milliampere and 1.0 amperes of 2 percent or better.
3.	DIGITAL VOLT-OHM-METER (DVOM)	With accuracy and resolution of ± 0.2 percent or better from 0.9 to 99 vdc and accuracy and resolution of 1 percent or better from 100 ohms to 100 kilohms. Input impedence of at least 10 megohms on all d-c voltage scales.
4.	OSCILLOSCOPE	Voltage and time accuracy and resolution of ± 5 percent; rise time 100 nanoseconds or less; sweep periods from 1 microsecond to 10 milliseconds per centimeter, amplitude scales from 0.1 to 10 volts/centimeter; with automatic triggering.

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5. ASSORTED COMPONENTS. Resistor - 2.5 megohms ± 10 percent, 2 watt.
Resistor - 10 kilohms ± 5 percent, 1/2 watt.
Resistor - Variable 0-1 megohm ± 10 percent, 2 watt.
Clip Lead - Less than 0.05 ohms.

4.5 Test methods and procedures

4.5.1 Environmental. As specified in MIL-C-60898 except the requirements of 3.3.1 herein shall be verified during temperature extremes.

4.5.2 Electrical test.

4.5.2.1 Diodes, reverse current. Connect the AMMETER, item 2 of 4.4.1.1, and the POWER SUPPLY, item 1 of 4.4.1.1, per Figure 4 and apply to each pin listed in Table I.

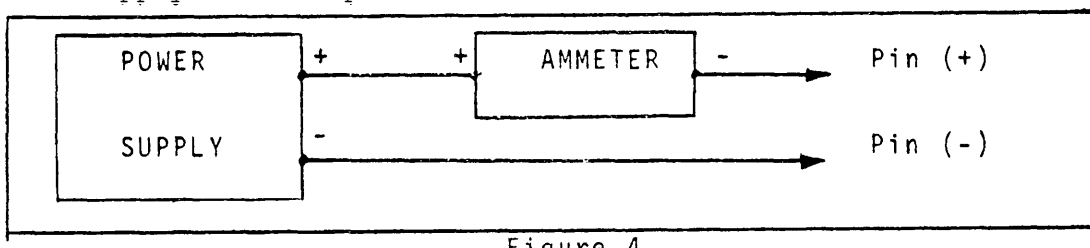


Figure 4

With the POWER SUPPLY voltaqe set at 28.0 ± 0.1 vdc, verify that the AMMETER reads 0.02 MILLIAMPERES or less for each connection Table I.

4.5.2.2 Diodes, forward drop. Connect the AMMETER, item 2 of 4.4.1.1, and the POWER SUPPLY, item 1 of 4.4.1.1, per Figure 4 and apply to each pin listed in Table II. With the AMMETER current set at 1.0 ± 0.1 ampere, verify, with the DVOM, item 3 of 4.4.1, that the voltage between each set of printed circuit boards pins is 0.9 ± 0.3 vdc.

4.5.2.3 Resistance. Connect the DVOM, item 3 of 4.4.1.1, between pin 21 and pin 27 and verify conformance to 3.4.3.

4.5.2.4 Pulse generator

4.5.2.4.1 High frequency. Using ASSORTED COMPONENTS, item 5 of 4.4.1.1, and POWER SUPPLY, item 1 of 4.4.1.1, make connections to the printed circuit assembly per Figure 1. R1 shall be 0.0 ohms (shorted) and the power supply (+V) shall be 22.0 ± 0.1 vdc.

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Connect the OSCILLOSCOPE, item 4 of 4.4.1.1 between pin 15 and pin 28 by adjusting R2, the waveform as observed on the OSCILLOSCOPE shall conform to the requirements of 3.4.4.1.

4,5. 2,4.2 Low frequency. With the connections of 4.5.2.4.1, change R1 to 2.5 megohms ± 10 percent (unshorted) and power supply (+V) to 28.0 ± 0.1 vdc. By adjusting R2, the waveform as observed on the OSCILLIOSCOPE shall conform to the requirements of 3.4.4.2.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Packaging Data Sheet 10554186.

6. NOTES

6.1 The notes specified in MIL-C-60898 are applicable to this specification.

Custodian:

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Preparing Activity:

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